

Yeon Seonwoo

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yeonsw.github.io

RESEARCH INTERESTS

My research focuses on how intelligent systems acquire, preserve, compose, and seek knowledge. I study the mechanisms through which knowledge is learned, maintained, connected, and actively acquired to solve complex tasks. My work spans continual learning, retrieval, large language models, reasoning, and agentic systems, with the goal of understanding how intelligent behavior emerges from the interaction between learned knowledge and external environments. I am particularly interested in building systems that can obtain the knowledge they need to reason and act effectively under real-world constraints such as latency, scalability, and reliability.

KEYWORDS

Large Language Models (LLMs), LLM Pre-training, Inference Optimization, Agentic Search, Retrieval-Augmented Generation (RAG), Query Understanding

EXPERIENCE

Applied Scientist, Amazon (Manhattan, NY)

Jul 2023 - Current

- **Knowledge Acquisition and Preservation in Foundation Models:** Led development of continual pre-training methodologies for small- to mid-sized language models, studying how domain-specific knowledge can be incorporated into foundation models while preserving previously acquired knowledge and instruction-following capabilities. This work investigated fundamental challenges in continual adaptation, including catastrophic forgetting, retention and utilization of acquired knowledge, and degradation of instruction-following behavior, and enabled broad adoption of domain-specialized language models across Amazon shopping applications.
- **Knowledge-Grounded Reasoning for Search Systems:** Developed entity-grounded query understanding systems for Prime Video Search that connect language-model reasoning with large-scale catalog knowledge. To support real-time deployment under strict latency constraints, I developed inference optimization algorithms based on speculative decoding and multi-token prediction, enabling reliable and efficient knowledge-grounded generation.

Applied Scientist Intern, Amazon (Bellevue, WA)

Mar 2022 - Jun 2022

Developed an unsupervised sentence representation learning method to more effectively capture semantic similarity between texts.

Research Scientist Intern, Adobe Research

Sep 2021 - Dec 2021

Developed word- and relation-level semantic graph representations of documents for domain-specific document retrieval.

EDUCATION

KAIST

Daejeon, Republic of Korea

Ph.D. Student in School of Computing

Sep 2017 - Aug 2022

Advisor: Alice Oh

Thesis title: "Distant-Supervision for Question Answering"

KAIST

Daejeon, Republic of Korea

M.S. in School of Computing

Mar 2016 - Aug 2017

Advisor: Alice Oh

Thesis title: "Event Prediction Using Vector Representation in Hawkes Processes"

Sungkyunkwan University

Suwon, Republic of Korea

B.S. in Computer Science & Engineering

Mar 2012 - Feb 2016

AWARDS

Naver Ph.D. Fellowship, 2020

- PUBLICATION** Juhee Son, **Yeon Seonwoo**, Alice Oh, James Thorne, Seunghyun Yoon, “Multi-hop Database Reasoning with Virtual Knowledge Graph”, *Workshop on Knowledge Graphs and Large Language Models 2024*.
- Chani Jung, Dongkwan Kim, Jiho Jin, Jiseon Kim, **Yeon Seonwoo**, Yejin Choi, Alice Oh, Hyunwoo Kim, “Perceptions to Beliefs: Exploring Precursory Inferences for Theory of Mind in Large Language Models”, *EMNLP 2024*.
- Yeon Seonwoo**, Guoyin Wang, Sajal Choudhary, Changmin Seo, Jiwei Li, Xiang Li, Puyang Xu, Sunghyun Park, Alice Oh. “Ranking-Enhanced Unsupervised Sentence Representation Learning.” *ACL 2023*
- Yeon Seonwoo**, Seunghyun Yoon, Franck Dernoncourt, Trung Bui, Alice Oh. “Virtual Knowledge Graph Construction for Zero-Shot Domain-Specific Document Retrieval.” *COLING 2022*
- Changyoon Lee, **Yeon Seonwoo**, and Alice Oh. “CS1QA: A Dataset for Assisting Code-based Question Answering in an Introductory Programming Course.” *NAACL 2022*
- Yeon Seonwoo**^{*}, Juhee Son^{*}, Jiho Jin, Sang-Woo Lee, Ji-Hoon Kim, Jung-Woo Ha, and Alice Oh. “Two-Step Question Retrieval for Open-Domain QA.” *ACL-Findings. 2022*.
- ^{*}Contributed equally
- Yeon Seonwoo**, Sang-Woo Lee, Ji-Hoon Kim, Jung-Woo Ha, and Alice Oh. “Weakly Supervised Pre-Training for Multi-Hop Retriever.” *ACL-Findings. 2021*.
Code URL: <https://github.com/yeonsw/LOUVRE>
- Yeon Seonwoo**, Ji-Hoon Kim, Jung-Woo Ha, and Alice Oh. “Context-Aware Answer Extraction in Question Answering.” *EMNLP (long). 2020*.
Code URL: <https://github.com/yeonsw/BLANC>
- Yeon Seonwoo**, Sungjoon Park, Dongkwan Kim, and Alice Oh. “Additive Compositionality of Word Vectors.” *Workshop on Noisy User-generated Text at EMNLP. 2019*.
Code URL: <https://github.com/yeonsw/OLIVE>
- Sungjoon Park, **Yeon Seonwoo**, Jiseon Kim, Jooyeon Kim and Alice Oh. “Denoising Recurrent Neural Networks for Classifying Crash-Related Events.” *IEEE Transactions on Intelligent Transportation Systems. 2019*.
- Yeon Seonwoo**, Sungjoon Park, and Alice Oh. “Hierarchical Dirichlet Gaussian Marked Hawkes Process for Narrative Reconstruction in Continuous Time Domain.” *EMNLP (long). 2018*.

- INVITED TALKS** **Semantic Alignment at Various Context Sizes**
BrainLink: State, Limitations, and Future of Large Language Models (LLM), South Korea Oct. 26, 2022
- Weakly-Supervised Pre-Training for Multi-Hop Retrieval**
KISTI, South Korea Feb. 08, 2022
Naver, South Korea Dec. 22, 2021
LG AI Research, South Korea Nov. 22, 2021
- Context-Aware Answer Extraction in Question Answering**
Naver, South Korea Jul. 01, 2020
- Additive Compositionality of Word Vectors**
Naver, South Korea Dec. 14, 2019
- Hierarchical Dirichlet Gaussian Marked Hawkes Process for Narrative Re-**

construction in Continuous Time Domain
NCSoft AI, South Korea

Jan. 25, 2019

**ACADEMIC
SERVICE**

Reviewer

ACL Rolling Review (Oct 2021, Nov 2021, Jan 2022, Apr 2022, June 2022, July 2022, Sep 2022), NeurIPS2022, EMNLP 2022, ICLR 2022, NeurIPS 2021, EMNLP 2021, ACL 2021, ICLR 2021

SCHOLARSHIP

Korea National Science&Technology Scholarship (Mar 2012 - Feb 2016)

REFERENCES

Prof.Alice Oh, School of Computing, KAIST, alice.oh@kaist.edu